9.7 Binomial Distributions

Name _____ Period ____ Date _____

Algebra I 9.7 · Student sheet

I can…

- make and read a probability distribution.
- tell when an experiment is binomial.
- find binomial probabilities and interpret a binomial distribution.

Vocabulary · fill in the missing word

Random variable — a variable whose value is _____ by the outcome of an experiment

Probability distribution — a table, graph, or rule that gives the probability of each value of a random _____; the probabilities add to 1

Binomial experiment — n _____ trials, each with two outcomes (success or failure), and the same probability p of success each time

Binomial probability — P(k _____) = $_nC_k \cdot p^k \cdot (1 - p)^{n-k}$

Part 1 · Four coins

List all 16 outcomes of flipping 4 coins (HHHH, HHHT, …). Sort them by the number of heads.

Heads	0	1	2	3	4
Outcomes					
Probability					
Your 16 flips					

Make a histogram of the probabilities on the back. Which number of heads is most likely? Why?

Part 2 · Notes: fill in the blanks

In a probability distribution, the probabilities add to _____.

A **binomial experiment** has n _____ trials, each with _____ outcomes and the same probability p.

P(k successes) = _____

For a fair coin: P(k heads in n flips) = _____ ÷ _____

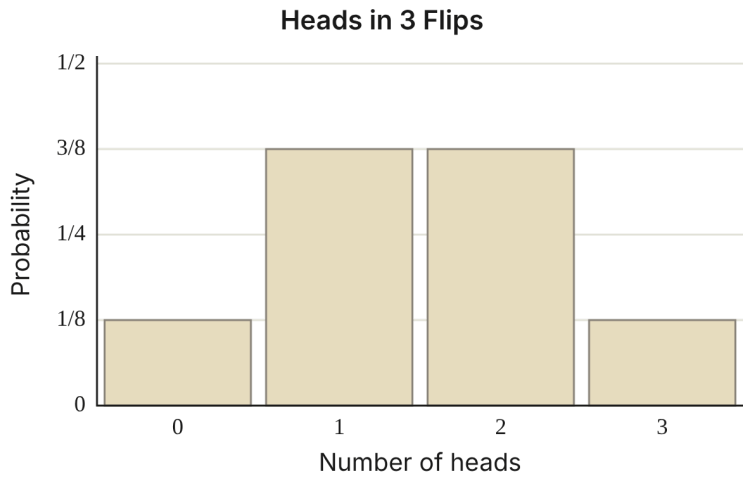
Part 3 · Examples (work them with the class)

Example 1

A bag has 3 balls labeled 1, 2 balls labeled 2, and 1 ball labeled 3. Let x be the number on a ball chosen at random. Make the probability distribution.

Example 2

The histogram shows the distribution of x, the number of heads in 3 coin flips. What is the probability of getting at least 2 heads?



Example 3

You flip a coin 10 times. What is the probability of getting exactly 7 heads?

Example 4

At a school, 70% of the students ride the bus. You choose 5 students at random. What is the probability that exactly 3 of them ride the bus?

Part 4 · Partner practice

Partner 1

You flip 6 coins. What is the probability of getting exactly 2 heads?

Partner 2

A basketball player makes 80% of her free throws. She shoots 4. Find the probability that she makes all 4, and the probability that she makes exactly 3.